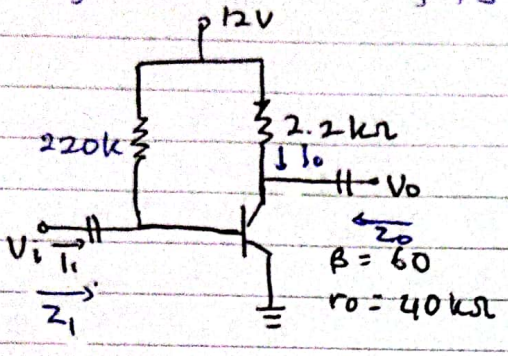
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~LATIHAN SOAL BAB 8 - BJT Small-Signal Analysis

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NIM : 211331037
Kelas : 1B Teknik Telekomunikasi

1. for the network of fig. 8.64



a). Zi dan Zo

$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{22k\Omega}$

$I_B = 51.36\ \mu A$

$I_E = I_B(\beta+1) = 61 \cdot 51.36\ \mu A = 3.13\ mA$

$r_e = \frac{26mV}{I_E} = \frac{26\ mV}{3.13mA} = 8.31\Omega$

$Z_1 = R_B // \beta r_e = 220k // (60)(8.31)$

$= 22k // 490.6$

$= 497.47\ \Omega$

$Z_o = R_L = 2.2k\Omega.$

b). Av & Ai

$Av = -\frac{R_c}{r_e} = \frac{-2.2k}{8.31} = -264.74$ $\quad Ai = \beta = 60$

c). Repeat part (a) with $r_o = 20k\Omega$

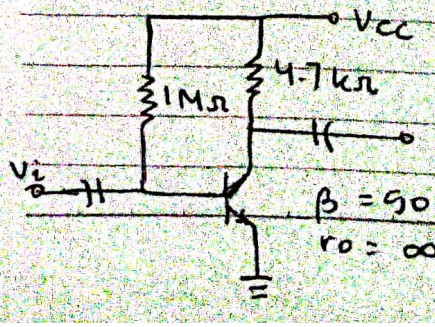
$Z_1 = 497.47\ \Omega$

$Z_o = r_o // R_c = 20k // 2.2k = \frac{20 \cdot 2.2}{20+2.2} = 1.98k\Omega$

d). Repeat part (b) with $r_o = 20k\Omega$

- $Av = \frac{-R_c // r_o}{r_e} = \frac{-1.98k}{8.31} = -238.27$ //
- $Ai = -Av \frac{Z_i}{R_c} = -(-238.27)\frac{(497.47)}{2.2k} = 53.88$ //

2. for the network of fig. 8.65, determine Vcc for a voltage gain of $Av = -200$.



$r_e = \frac{-R_c}{Av} = \frac{-4.7k}{-200} = 23.5\Omega$

$I_E = \frac{26mV}{r_e} = \frac{26mV}{23.5} = 1.106\ mA$

$I_B = \frac{I_E}{\beta+1} = \frac{1.106\ mA}{91} = 12.15\ \mu A.$

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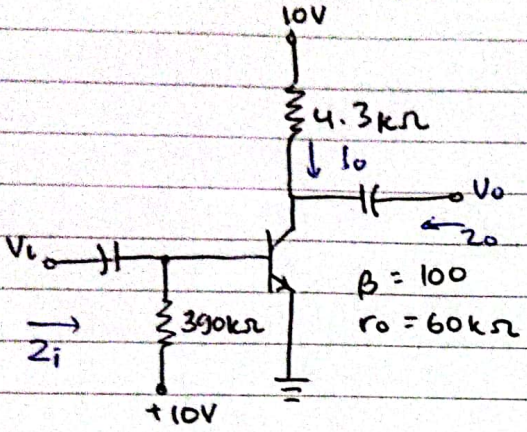
$V_{CC} = I_B R_B + V_{BE}$

$= (12.15\ \mu A)(1M) + 0.7$

$= 12.15V + 0.7$

$= 12.85\ V$

3. for the network of fig. 8.66



a). I_B, I_C dan re

$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{10 - 0.7}{390k} = 23.85\ \mu A$

$I_E = (\beta + 1) I_B = (101)(23.85\ \mu A)$

$= 2.41\ mA$

$r_e = \frac{26mV}{I_E} = \frac{26\ mV}{2.41\ mA} = 10.79\ \Omega$

$I_C = \beta I_B = (100)(23.85\ \mu A)$

$= 2.38\ mA$

b). Zi dan Zo

$Z_i = R_B // \beta r_e = 390k // 100\ (10.79\ \Omega) = 390k // 1079$

$= 1.08\ k\Omega$

$Z_o = R_C = 4.3\ k\Omega$

c). Av dan Ai

$A_V = \frac{-R_C}{r_e} = \frac{-4.3k}{10.79\Omega} = -348,52$, $A_i = \beta = 100$

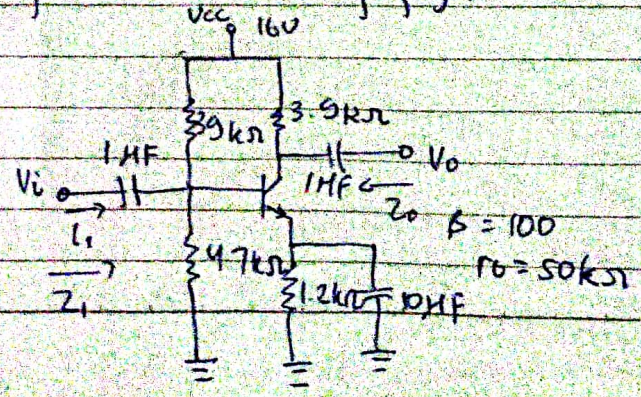
$A_i = \beta = 100$

d). $r_o = 30k\Omega$ → Av dan Ai ?

$A_V = \frac{-R_C // R_o}{r_e} = \frac{-4.3k // 30k}{10.79} = -348.47$

$A_i = A_V \frac{Z_i}{R_C} = -348.47 \frac{1.08k}{4.3k} = 87.52$

4. for the network of fig.



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a). re

$V_B = \frac{R_2 \cdot V_{CC}}{R_1 + R_2} = \frac{4.7k}{39k+4.7k}\,16 = 1.721\ V$

$V_E = V_B - V_{BE} = 1.721 - 0.7 = 1.021\ V$

$I_E = \frac{V_E}{R_E} = \frac{1.021\ V}{1.2k} = 0.85\ mA$

$r_e = \frac{26\ mV}{0.85\ mA} = 30,56\ \Omega$

b). Z_i dan Z_o

$Z_i = R_1 \,//\, R_2 \,//\, \beta r_e$

$= 39k \,//\, 4.7k \,//\, (100)(30.56)$

$Z_i = 1.768\ k\Omega$

$Z_o = R_C = 3.9\ k\Omega$

c). Av dan Ai

$A_V = \frac{-R_C}{r_e} = \frac{-3.9k\Omega}{30,56\Omega} = -126.6$

$A_i = -A_V \frac{Z_i}{R_C} = (-127,6)\cdot\frac{1.768k\Omega}{3.9\Omega} = 57.85$

d). Ulangi (b) dan (c) dg $r_o = 25k$

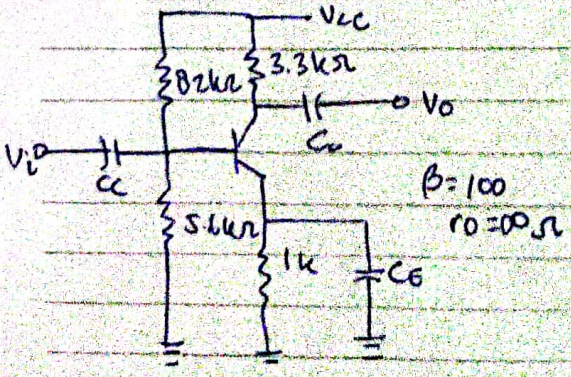
→ $Z_i = 1.768k\Omega$

$Z_o = R_C \,//\, r_o = 3.9k \,//\, 25k = 3.37k\Omega$

→ $A_V = \frac{-R_C \,//\, r_o}{r_e} = \frac{-39k \,//\, 25k}{30.56} = -110,28$

$A_i = -A_V \cdot \frac{Z_i}{R_C} = (-110.28)\cdot\frac{1,768k}{3.9k} = 50$

5 Determine VCC for the network of Fig. IF $A_V = -160$ and $r_o = 100k\Omega$



$\beta = 100$

$r_o = 100\Omega$

- $r_e = \frac{-R_C}{A_V} = \frac{-3.3k\Omega}{-160} = 20.625\Omega$
- $I_C = \frac{26mV}{r_e} = \frac{26\ mV}{20.625\Omega} = 1.261\ mA$
- $V_E = I_E R_E = 1.261\ mA \cdot 1k\Omega = 1.261\ V$
- $V_B = V_{BE} + V_E = 0.7 + 1.261V = 1.961\ V$
- $V_B = \frac{5.6k \cdot V_{CC}}{5.6k+82k} \Rightarrow 1,961 \cdot \frac{5.6k}{27.6k}$

→ $V_{CC} = \frac{82.6k \cdot 1961}{5.6k}$

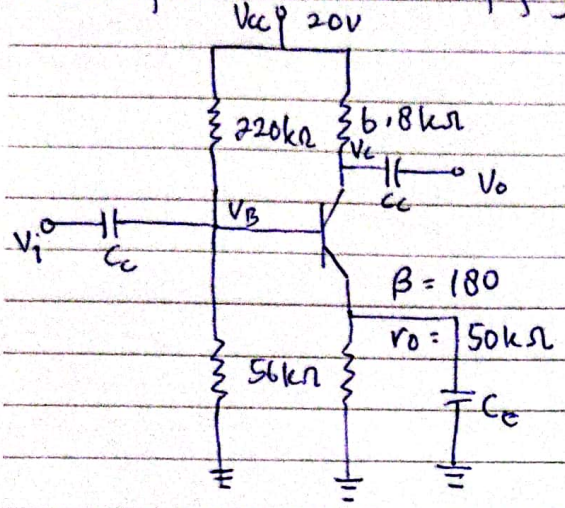
$V_{CC} = 30.68\ V$

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6 for the network of fig. 8.69



a). re

- $R_{TH} = 56k \,//\, 220k = 44,64\,k\Omega$
- $E_{TH} = \frac{56k \cdot 20}{220k+56k} = 4.058\ V$
- $I_B = \frac{E_{TH} - V_{BE}}{R_{TH}+(\beta+1)R_E} = \frac{4.058 - 0.7}{44.64k + (181)(2.2k)} = 7.58\ \mu A$
- $I_E = (\beta+1)(I_B) = 181\,(7.58\ \mu A) = 1.372\ mA$

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{1.372mA} = 18.95\ \Omega$$ //

b). V_B dan V_C

$V_E = I_B \cdot R_E = 1.372\ mA \cdot 2.2k\Omega$
$= 3.02V$

$V_B = V_E + V_{BE} = 3.02 + 0.7 = 3.72V$

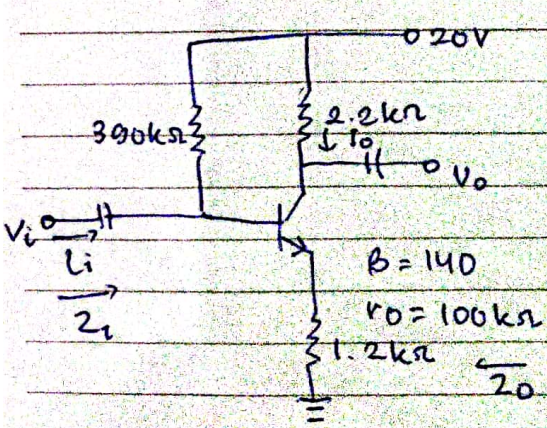
$V_C = V_{CC} - I_C R_C$
$= 20 - \beta i_b R_C = 20 - (180)(7.58\ \mu A)(6.8k$
$= 10.72V$

c). Zi dan $Av = V_o / V_i$

- $Z_i = R_1 \,//\, R_2 \,//\, \beta r_e$
  $= 56k \,//\, 220k \,//\, 180.18.95k$
  $= 44.64k \,//\, 3.41k$
  $Z_i = 3,17k\Omega$

- $Av = \frac{-R_C \,//\, r_o}{r_e} = \frac{-6.8k \,//\, 50k}{18.95\Omega}$
  $= -315,88$

7 for the network of fig.



a). re

$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{20 - 0.7}{390k + (141)1.2k}$
$= 34.51\ \mu A$

$I_e = (\beta+1)I_B = (141)(34.51\ \mu A)$
$= 4.866\ mA$

$r_e = \frac{26\ mV}{4.866mA} = 5.34\ \Omega$

b). Zi dan Zo

$Z_b = \beta r_e + (\beta+1)R_E$
$= (140)(5.34) + (141)1,2k$
$Z_b = 169.95k\Omega$

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- $Z_i = R_B // Z_b = 390k // 164.95k = 118.37k\Omega$
- $Z_o = R_C = 2.2k\Omega$

c). Av dan Ai

$A_v = \frac{-\beta R_c}{Z_b} = \frac{-140 \cdot 2.2k}{164.95k} = -1.81$ $\quad A_i = -A_v \cdot \frac{Z_i}{R_c} = -(-1.81) \cdot \frac{118.37k}{2.2k}$

$= 97.39.$

d). Ulangi (a) & (b) $r_o = 20k\Omega$

- $Z_b = \beta r_e : \left\{ \frac{(\beta+1) R_c // r_o}{1 + \frac{(R_c + R_b)}{r_o}} \right\} \cdot r_e = 747.6 \left\{ \frac{141 + 2.2k // 20k}{1 + 3.4k // 20k} \right\} \cdot 1.2k$

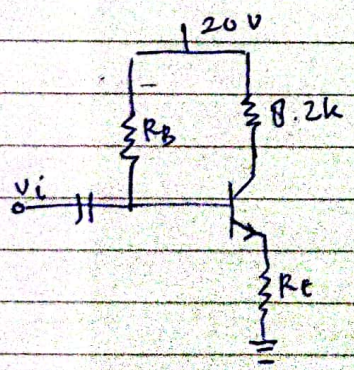
$Z_b = 145.47\ k\Omega$

- $Z_i = R_b // Z_b = 390k // 145.47k$
  $= 105.95k\Omega$
- $Z_o = R_C = 2.2k\Omega$
- $A_v = \frac{V_o}{V_i} = \frac{\frac{-\beta R_c}{Z_b}\left(1 + \frac{r_e}{r_o}\right) + \frac{R_c}{r_o}}{1 + \frac{R_c}{r_o}} = \frac{-140 \cdot 2.2k}{145.47k} = \frac{\left(1 + \frac{5.34}{20k}\right) + \frac{2.2k}{20k}}{1 + \frac{2.2k}{20k}}$

$A_v = -1.01$

- $A_i = -A_v \cdot \frac{Z_i}{R_c} = -(-1.81) \cdot \frac{105.95k}{2.2k} = 87.17$ //

[8] for the network of fig 8.71, determine R_E and R_B if $A_v = -10$ and $r_e = 3.8\Omega$. Assume that $Z_b = \beta r_e$.



$R_E = \frac{R_c}{10} = \frac{8.2k}{10} = 820\Omega$ //

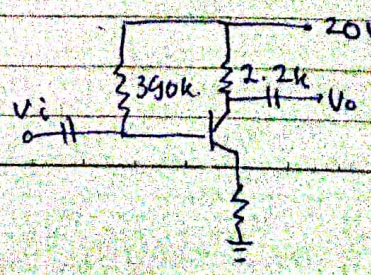
$I_E = \frac{26mV}{3.8} = 6.482\ mA$

- $V_E = I_e R_E = (6.482\ mA)(820\Omega) = 5.61\ V$
- $V_B = V_E + V_{BE} = 5.61 + 0.7 = 6.31\ V$
- $I_B = \frac{I_E}{\beta+1} = \frac{6.842}{121} = 56.55\ \mu A$
- $R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_B}{I_B} = \frac{20 - 6.31}{56.55\ \mu A} = 242.09\ k\Omega$ //

[9] Repeat problem 7 with R_E bypassed. Compare results.

a). dc analisis seperti no 7

$r_e = 5.34\ \Omega$



$\beta = 140$

$r_o = 100k\Omega$

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b). Z_i dan Z_o

$Z_i = R_B // Z_b = R_b // \beta r_e$

$= 390k // 140 \cdot 5.34$

$Z_i = 764.17\,\Omega$

c). Av & Ai

$\cdot Av = \frac{-R_c}{r_e} = \frac{-2.2k}{5.34} = -411.99$ //

$\cdot Ai = \frac{\beta R_B}{R_B + Z_b} = \frac{140 \cdot 390k}{390k + 0.746k}$

$= 139.73$ //

d). $Z_i = 746.17\,\Omega$

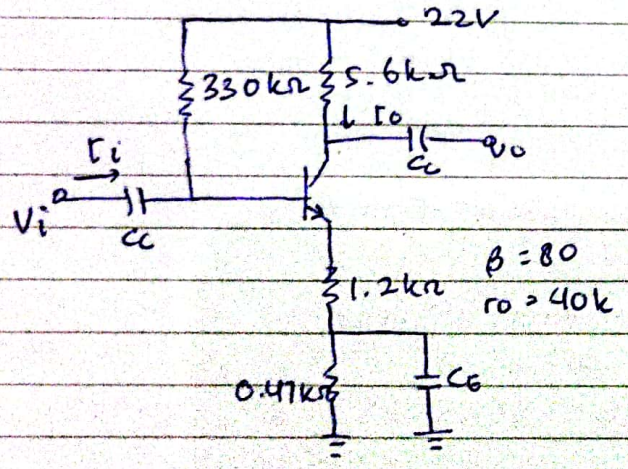
$Z_o = R_c // r_o = 2.2k\Omega // 20k$

$Z_o = 1.98\,k\Omega$

$\cdot Av = \frac{-R_c // r_o}{r_e} = \frac{-1.98}{5.34} = -370.8$

$\cdot Ai = -Av \cdot \frac{Z_i}{R_c} = -(-370.8) \cdot \frac{746.17}{2.2k} = 125.76$

10 for the network of fig.



a). re

$\cdot I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{22V - 0.7}{330k + 181(1.2k + 0.47k)}$

$= 45.78\,\mu A$

$I_E = (\beta+1)I_B = 81(45.78\,\mu A)$

$= 3.71\,mA$

$r_e = \frac{26mv}{3.71mA} = 7\,\Omega$

b). Z_i dan Av

$Z_b = \beta r_e + \left\{ \frac{(\beta+1) + R_c/r_o}{1 + (R_c + R_E)/r_o} \right\} R_E$

$80\,(70)\left(\frac{81 + 0.14}{1 + 0.17}\right) = 1.2k$

$Z_b = 83.73\,k\Omega$.

$\cdot Z_i = R_b // Z_b = 330k // 83.78k$

$= 66.82k\Omega$

$\cdot Av = \frac{\frac{\beta R_c}{Z_b}\left(1 + \frac{r_e}{r_o}\right) + \frac{R_c}{r_o}}{1 + \frac{R_c}{r_o}}$

$= \frac{-80\,(5.6k)}{93.78k\Omega}\left(1 + \frac{7}{40k}\right) = \frac{5.6k}{40k}$

$Av = -4.57$

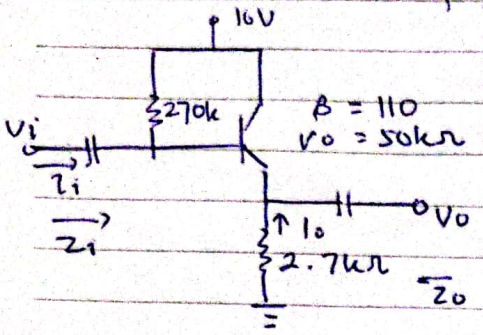
c). $Ai = -Av \cdot \frac{Z_i}{R_c} = -(-4.57) \cdot \frac{(66.82k)}{5.6k} = 54.53$.

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11. For the network of fig



a). re dan βre

$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{16 - 0.7}{270k + (111)\,2.7k} = 26.86\ \mu A$

$I_E = (\beta+1)I_B = (111)\,26.86\ \mu A = 2.98\ mA$

$r_e = \frac{26mV}{I_E} = \frac{26mV}{2.98mA} = 8.72\ \Omega$

$\beta r_e = (110)(8.72) = 959.2\ \Omega$

b). Zi dan Zo

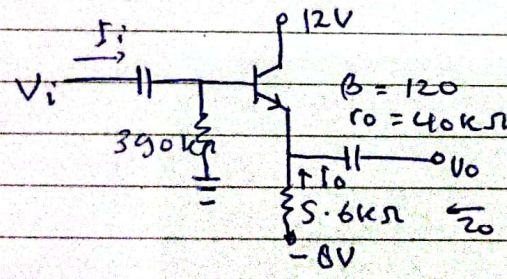
- $Z_b = \beta r_e + (\beta+1)R_E = 959.2 + (111)\,2.7k = 300.66\ k\Omega$
- $Z_i = R_B \,//\, Z_b = 270k \,//\, 300.66k = 142.25\ k\Omega$
- $Z_o = R_E \,//\, r_e = 2.7k \,//\, 8.72 = 8.69\ \Omega$

c). Av & Ai

- $A_V = \frac{R_E}{R_E + r_e} = \frac{2.7k}{2.7k + 8.72} = 0.997$ //

$A_i = -A_V \frac{Z_i}{R_L} = -(0.997)\frac{142.25k}{2.7k} = -52.53$ //

12. for the network of fig



a). Zi dan Zo

$I_B = \frac{V_{CE} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{8V - 0.7V}{390k + (121)5.6k\Omega} = 6.84\ \mu A$

$I_E = (\beta+1)I_B = (121)(6.84\ \mu A) = 0.82\ mA$

$r_e = \frac{26mV}{I_E} = \frac{26mV}{0.828\ mA} = 3.14\Omega$

$r_o > 10R_E$

$Z_b = \beta r_e + \frac{(\beta+1)R_E}{1 + R_E/r_o}$

$= (120)(31.4\Omega) + \frac{(121)(5.6k\Omega)}{1 + 5.6k\Omega/40k\Omega}$

$= 3.77\ k\Omega + 594.39\ k\Omega$

$= 598.16k\Omega$

$Z_i = R_B \,//\, Z_b = 390k\Omega \,//\, 598.16k\Omega$

$= 236.1k\Omega$

$Z_o = R_E \,//\, r_e = 5.6k\Omega \,//\, 31.4\Omega$

$= 31.2\ \Omega$

b). $A_V = \frac{(\beta+1)R_E / Z_b}{1 + R_E/r_o}$

$= \frac{(121)(5.6k\Omega / 598.16k\Omega}{1 + 5.6k\Omega / 40k\Omega}$

$= 31.2\ \Omega$

c). Vo if Vi = 1 mV

$A_V = \frac{V_o}{V_i} = 0.994$

$V_o = A_V V_i = (0.994)(1mV)$

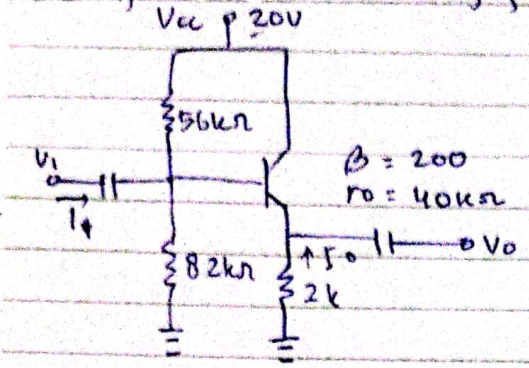
$= 0.994\ mV$

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13 for the network of frg.



a). I_B dan I_C

$$V_B = \frac{8.2k \cdot 20}{8.2k + 56k} = 2.5545V$$

$$V_E = V_B - V_{BE} = 2.5545 - 0.7 = 1.855\,V$$

$$I_E = \frac{V_E}{R_E} = \frac{1.855}{2k} = 0.927\,mA$$

$$I_B = \frac{I_E}{(\beta+1)} = \frac{0.927\,mA}{201} = 4.61\,\mu A$$

$$I_C = \beta I_B = (200)\,4.61\,\mu A = 0.922\,mA$$

b). $r_e = \frac{26mV}{I_E} = \frac{26mV}{0.927\,mA} = 28.05\,\Omega$

c). Z_i dan Z_o

$$Z_b = \beta r_e + (\beta+1) R_E$$
$$= (200)\,28.05 + (201)\,2k$$
$$Z_b = 407.16\,k\Omega$$

$$Z_i = 56k \,//\, 8.2k \,//\, 407.61k$$
$$= 7.15k \,//\, 407.61k$$
$$Z_i = 7.03\,k\Omega$$

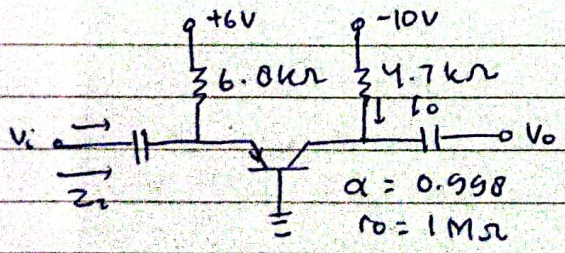
$$Z_o = R_E / r_e = 2k \,//\, 28.05 = 27.66\,\Omega$$

d). Av dan Ai

$$A_V = \frac{R_E}{R_E + r_e} = \frac{2k}{2k + 28.05} = 0.986$$

$$A_i = -A_V \cdot \frac{Z_i}{R_E} = -(0.986)\,\frac{7.03k}{2k}$$
$$= -3.47$$

14



a). re

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6V - 0.7V}{6.8k\Omega}$$
$$= 0.779\,mH$$

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{I_E} = \frac{26mV}{0.779mA}$$
$$= 33.38\,\Omega$$

b). find Z_i dan Z_o

$$Z_i = R_E \,//\, r_e = 6.8k\Omega \,//\, 33.38\,\Omega$$
$$= 33.22\,\Omega$$

$$Z_o = R_C = 4.7\,k\Omega$$

c). Av dan Ai

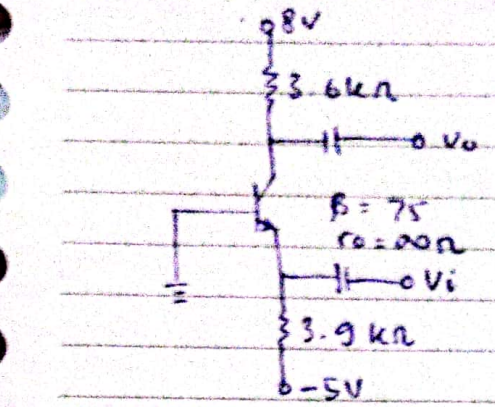
$$A_V = \frac{\alpha R_C}{r_e} = \frac{(0.998)(4.7k\Omega)}{33.38\,\Omega} = 140.52$$

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15 for the network of fig. 8.77, determine A_v dan A_i



$$\alpha = \frac{\beta}{\beta+1} = \frac{75}{76} = 0.9868$$

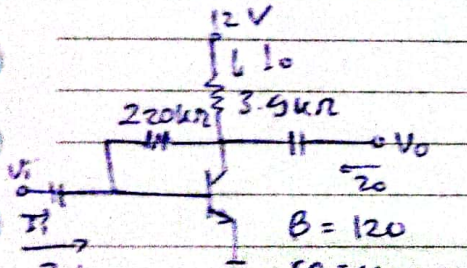
$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{5V - 0.7V}{3.9k\Omega} = 1.1\,mA$$

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{1.1mA} = 23.58\,\Omega$$

$$A_V = \alpha \frac{R_C}{r_e} = \frac{(0.998)(3.9k\Omega)}{23.58\Omega} = 163.2$$

$$A_i = \frac{I_o}{I_i} = \alpha \frac{I_E}{I_C} \quad \alpha = 0.9868$$

16 for the collector FB configuration of fig. 8.78



a). re

$$I_B = \frac{V_{CC} - V_{BE}}{R_E + \beta R_C} = \frac{12V - 0.7V}{220k\Omega + 120(3.9k)} = 16.42\,\mu A$$

$$I_E = (\beta+1) I_B = (120+1)(16.42\,\mu A) = 1.987\,mA$$

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{1.987mA} = 13.08\,\Omega$$

b). Zi dan Zo

ZI/HI/BFE/MIR

$$A_V = \frac{-R_C}{r_e} = \frac{-3.9k\Omega}{1.987mA} = -298$$

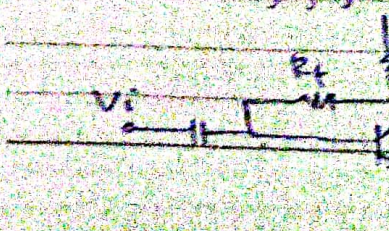
- $Z_i = \beta r_e \,//\, \frac{R_E}{A_V} = 120.13.08 \,//\, \frac{220k}{298} = 501,98\,\Omega$
- $Z_o = R_C // R_E = 3.9k // 220k = 3.83k\Omega$

c). Av dan Ai

$A_V = -298$

$$A_i = -A_V \frac{Z_i}{R_C} = (-298)\frac{501,98}{3.9k} = 38.37$$

17 Given $r_e = 10\Omega$, $\beta = 200$, $A_V = -160$ and $A_i = 19$ for the network of fig. 8.79 determine R_C, R_f and V_{CC}



$$R_C = 160 \cdot r_e = 160 \cdot 10\Omega = 1.6k\Omega$$

$$A_i = \frac{\beta r_e}{R_E + \beta r_e} \Rightarrow 19 = \frac{200 r_e}{R_B + 200 + 1.6k}$$

$$\rightarrow R_E = 33.59\,k\Omega$$

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$I_B = \frac{V_{CC} - V_{BE}}{R_E + \beta R_C} = I_B(R_E + \beta R_C) = V_{CC} - V_{BE}$

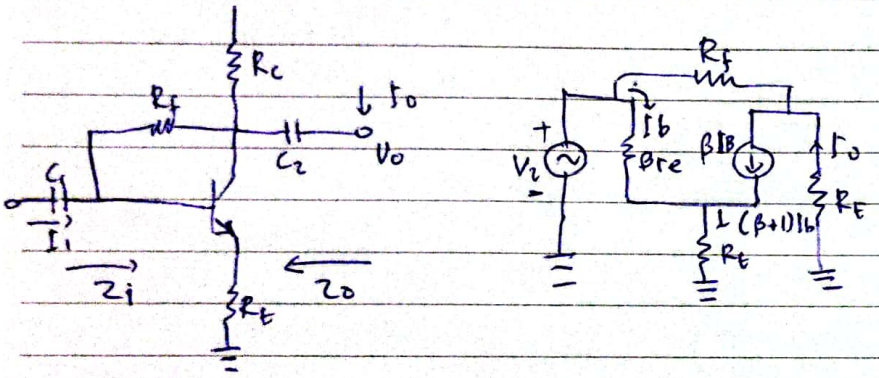
$V_{CC} = V_{BE} + I_B(R_E + \beta R_C)$

$\rightarrow I_E = \frac{26mV}{10\Omega} = 2.6mA$

$\rightarrow I_B = \frac{I_E}{\beta+1} = \frac{2.6\,mA}{201} = 12.94\,\mu A$

$V_{CC} = 0.7 + 12.94\,\mu A\,(33.59k + (200)1.6k\Omega) = 5.20V$

18 for the network of fig. 8.30



a). Av

$V_i = I_b\beta r_e + (\beta+1)I_b R_b$

$I_o + I' = I_c = \beta I_b$

$I_B = I' + I_b$

$I' = I_1 - I_b$

$I_o + (I_1 - I_b) = \beta I_b$

$I_o = (\beta+1)I_b - I_1$

$I_o = (\beta+1)I_b$

$V_o = -I_o R_c = -(\beta+1)I_b R_c$

$A_V = \frac{V_o}{V_i} = \frac{-R_c}{R_E}$

b). Ai, Kirchoff =

$V_i - V_{Rf} - V_o = 0$

$(I_b\beta r_e) + ((\beta+1)I_{RE}) - (I'R_f) - (I_o R_c) = 0$

$I'R_f = (I_1 - I_b)R_f = I_i R_f - I_B R_f$

$I_o = (\beta+1)I_b$

$I_b\beta r_e + \beta I_b R_E - I_i R_f = I_b R_f = I_o R_c = 0$

$I_o r_e + I_o R_E = I_1 R_f + \frac{\beta I_b}{\beta} R_f = I_o R_c = 0$

$I_o\left(r_e + R_E + \frac{R_f}{\beta} R_c\right) = I_i R_f$

$A_i = \frac{I_o}{I_1} = \frac{R_f}{R_E + R_C + R_E}$

c). Zi dan Zo

$V_i = \beta I_b(r_e + R_f)$

$= I' + I_b$

$= \frac{V_i - V_o}{R_f} + I_b$

$I_i = \frac{V}{R_f} + I_b$

$I_b = I_i + \frac{V_o}{R_f}$

$\rightarrow V_i = \beta R_E I_i + \beta \frac{V_o}{R_f} \cdot R_E$

$V_i = \beta R_E I_i + \frac{\beta A_i V_i R_E}{R_F}$

$V_i = \frac{A_V \beta R_f V_i}{R_f} = \beta R_E I_1$

$V_i\left(1 - \frac{A_V \beta R_E}{R_f}\right) = (\beta R_E) I_1$

$Z_i = \frac{V_i}{I_i} = \frac{\beta R_E}{1 - A_V \beta R_E} = \frac{\beta R_E R_f}{R_f}$

$Z_i = \frac{V_i}{I_i} = \frac{\beta R_E R_f}{\beta R_E / A_V + R_f}$

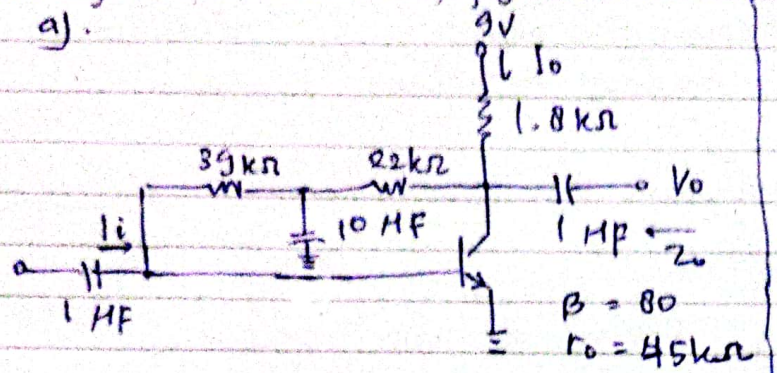
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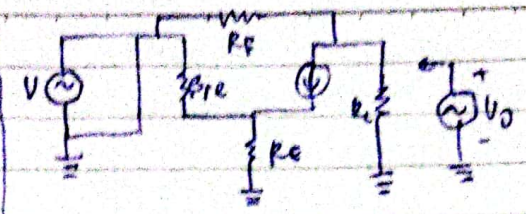
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19 for the network of fig. 8.80

a).





$V_i = I_b \cdot \beta r_e + (\beta+1) I_b R_E$

$V_i = \beta I_b (r_e + R_E)$; 0

$I_o = \frac{V_o}{R_c} + \frac{V_o}{R_f} = V_o \left(\frac{1}{R_c} + \frac{1}{R_f}\right)$

$Z_o = \frac{V_o}{I_o} = \frac{1}{\frac{1}{R_c} + \frac{1}{R_f}} = \frac{R_c \cdot R_f}{R_c + R_f} = R_c // R_f$

g). Z_i dan Z_o

$I_B = \frac{V_{cc} - V_{BE}}{R_f + \beta R_E} = \frac{9 - 0.7}{39k + 22k + 80(1.8k)}$

$= 40.49\ \mu A$

$I_E = (\beta+1) I_B = (81)\ 40.49\ \mu A = 3.28\ mA$

$r_e = \frac{26\ mV}{I_E} = \frac{26\ mV}{3.28\ mA} = 7.93\ \Omega$

- $Z_i = R_f // r_e \beta = 39k // 80.7.93 = 0.62k\Omega$
- $Z_o = R_c // R_f = 1.8k // 22k = 1.66k\Omega$

b). Av dan Ai

- $A_v = -\frac{R_E}{r_e} = \frac{-R_c // R_f}{r_e} = \frac{1.8 // 22k}{7.93}$

$= -209.82$

- $A_i = -\frac{A_v Z_i}{R_c} = \frac{-(209.82)(0.62k)}{1.8k}$

$= 72.27$

d). $R_c = 2.2k\Omega$ $R_E = 1.2k\Omega$

$R_f = 120k\Omega$ $\beta = 90$

- $A_i = -\frac{R_c}{R_E} = \frac{-2.2k}{1.2k} = -1.83$

$A_i = \frac{R_f}{R_E + R_c + R_f/\beta} = \frac{120k}{1.2k + 2.2k + 120k/90}$

$= 25.37$

$Z_o = R_c // R_f = 2.2k // 120k = 2.16k\Omega$

$Z_i = \frac{\beta R_E R_f}{\beta R_E / A_v R_f} = \frac{90 \cdot 1.2k \cdot 120k}{90 \cdot 1.2k (188) + 120k}$

$= 40.8\ k\Omega$

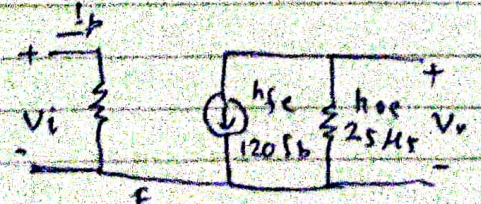
20 - Given $\beta = 120$, $r_e = 4.5\Omega$ and $r_o = 4k\Omega$, sketch the approximate equivalent circuit

- Given $h_{ie} = 1k\Omega$, $h_{re} = 2 \times 10^{-4}$, $h_{fe} = 90$, and $h_{oe} = 20\ \mu S$ sketch the re model

a). $\beta = 120$ $r_e = 4.5\Omega$ $r_o = 40k\Omega$

$h_{ie} = \beta r_e = 120 \cdot 4.5\Omega = 540\Omega$

$h_{oe} = \frac{1}{10} = \frac{1}{40k} = 20\ \mu S$



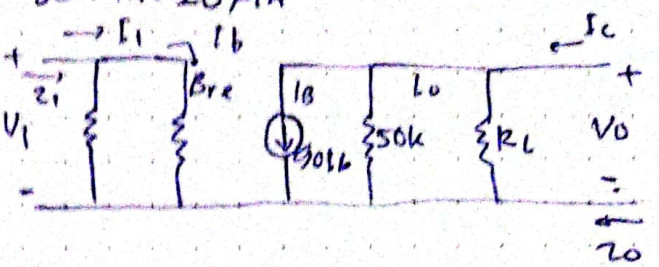
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b). hie = 1kΩ, hre = 2×10^{-4}, hfe = 90, hoe 20MA

• re = $\frac{hie}{\beta} = \frac{1k}{90} = 11.11\ \Omega$

• β = hfe = 90

$r_o = \frac{1}{hoe} = \frac{1}{20\mu S} = 50k\Omega$



21 for the network of problem 1.

a). Determine re = 8.31Ω

b). Hfe = β = 60

hie = βre = (60)(8.31Ω) = 498.6Ω

c). Zi dan Zo

$Av = \frac{-hfe\,Rc}{hie} = \frac{-(60)(2.2k\Omega)}{498.6\Omega} = -294.74$

Ai ≅ hfe = 60

d). Av dan Ai

$Av = \frac{-hfe\,Rc}{hie} = -\frac{(60)(2.2k\Omega)}{498.6\Omega} = -294.74$

Ai ≅ hfe = 60

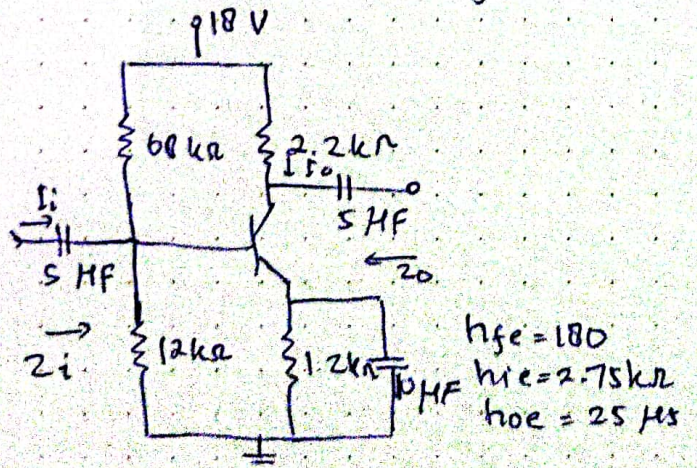
e). Zi dan Zo jika hoe = 50 μS

• Zi = 497.47 Ω (the same)

• Zo = ro // Rc, $ro = \frac{1}{25\mu S} = 40k\Omega \rightarrow 40k\Omega // 2.2k\Omega = 2.09k\Omega$

f). Av dan Ai hoe = 50 μS

22 for the network of fig 8.81



a). Zi dan Zo

68kΩ // 12kΩ = 10.2kΩ

• Zi = 10.2kΩ // hie = 10.2kΩ // 2.75k = 2.166kΩ

• Zo = Rc // ro = 2.2kΩ // 40kΩ = 2.085kΩ

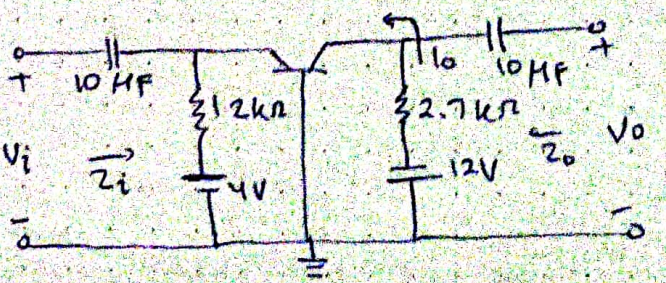
b). Av dan Ai

$Av = \frac{-hfe\,R'c}{hie}$ → R'c // ro = 2.085kΩ

$= \frac{-(180)(2.085k\Omega)}{2.75k\Omega} = -136.5$

$Ai = \frac{Io}{Ii} = \frac{Io}{I'i} \times \frac{I'i}{Ii} = 135.13$

23 for the common base network of fig.



hfb = −0.992

hib = 9.45Ω

hob = 1 MA/V

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a). Determine $Z_i = R_e // h_{ib} = 1.2k\Omega // 9.45\Omega = 9.38\Omega$

$Z_o = R_c // \frac{1}{h_{ib}} = 2.7k\Omega // \frac{1}{1\times10^{-6}A/V} = 2.7k\Omega // 1M\Omega \cong 2.7k\Omega$

b). $A_v = \frac{-h_{fb}(R_c // \frac{1}{h_{ab}})}{h_{ib}} = \frac{-(-0.992)(\cong 2.7k\Omega)}{9.45} = 284.43$

$A_i = -1$

c). Determine α, β, r_e and r_o.

$\alpha = -h_{fb} = -(-0.992) = 0.992$

$\beta = \frac{\alpha}{1-\alpha} = \frac{0.992}{1-0.992} = 124$

$r_e = h_{ib} = 9.45\Omega$

$r_o = \frac{1}{h_{ab}} = \frac{1}{1MA/V} = 1M\Omega$

24 Repeat parts (a) and (b) of problem 22 with $h_{re} = 2\times10^{-4}$ and compare result.

a). $Z_i' = h_{ie} = \frac{h_{fe}h_{re}R_L}{1+h_{oe}R_L}$

$= \frac{2.75k\Omega - (180)(2\times10^{-4})(2.2k\Omega)}{(1+25\mu S)(2.2k\Omega)}$

$= 2.60k\Omega$

$Z_i = 10.2k\Omega // Z_i' = 2.12k\Omega$

$Z_o' = \frac{1}{h_{oe}-(h_{fe}h_{re}/h_{ie})} = \frac{1}{25\mu S-(180)(2\times10^{-4})/2.75k\Omega}$

$= 83.75k\Omega$

$Z_o = 2.2k\Omega // 83.75k\Omega = 2.14k\Omega$

c). $A_i' = \frac{h_{fe}}{1+h_{oe}R_L} = \frac{180}{1+(25\mu S)(2.2k\Omega)} = 170.62$

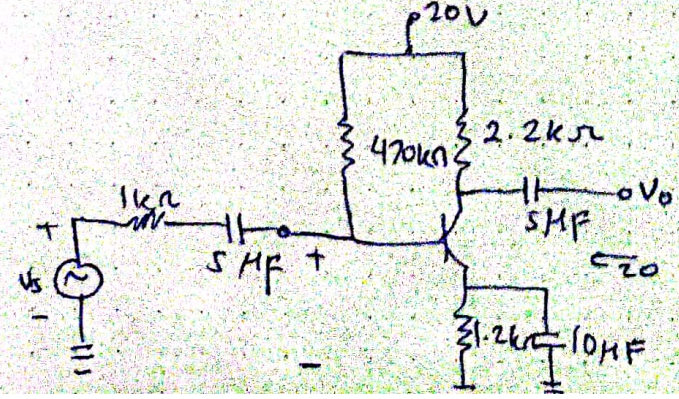
$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_i'}\cdot\frac{I_i'}{I_i} = (170.62)\left[\frac{10.2k\Omega}{10.2k\Omega+2.60k\Omega}\right]$

$= 135.13$

b). $A_v = \frac{-h_{fe}R_L}{h_{ie}+(h_{ie}h_{oe})-h_{fe}h_{re})R_L} = \frac{180)(2.2k\Omega)}{2.75k\Omega+(2.75k\Omega)(25\mu S)-(180)\cdot 2\times10^{-4}.22k}$

$= -140.3$ //

25 For the network of fig



$h_{fe} = 180$
$h_{ie} = 0.86k\Omega$
$h_{re} = 1.5\times10^{-4}$
$h_{oe} = 25\mu S$

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g). $Z_i' = h_{ie} = \frac{h_{fe} h_{re} R_L}{1 + h_{oe} R_L} = 0.86k\Omega - \frac{(140)(1.5\times10^{-4})(2.2k\Omega}{1 + (25\mu S)(2.2k\Omega)} = 0.86k\Omega - 43.79\Omega$

$= 816.21\Omega$ //

b). $A_V = \frac{-h_{fe} R_L}{h_{ie} + (h_{ie}h_{oe} - h_{fe}h_{re}) R_L} = \frac{-(140)(2.2k\Omega)}{0.86k\Omega + ((0.86k\Omega))(25\mu S) - (140)(1.5\times10^{-4}))2.2k\Omega}$

$= -357.68$

c). $A_i = \frac{I_o}{I_i} = \frac{h_{fe}}{1 + h_{oe} R_L} = \frac{140}{1 + (25\mu S)(2.2k\Omega)} = 132.70$

$A_i' = \frac{I_o}{I_i} = \left(\frac{I_o}{I_i}\right)\left(\frac{I_i}{I_i'}\right)$ - $I_i = \frac{470k\Omega I_i'}{470k\Omega + 0.818k\Omega}$

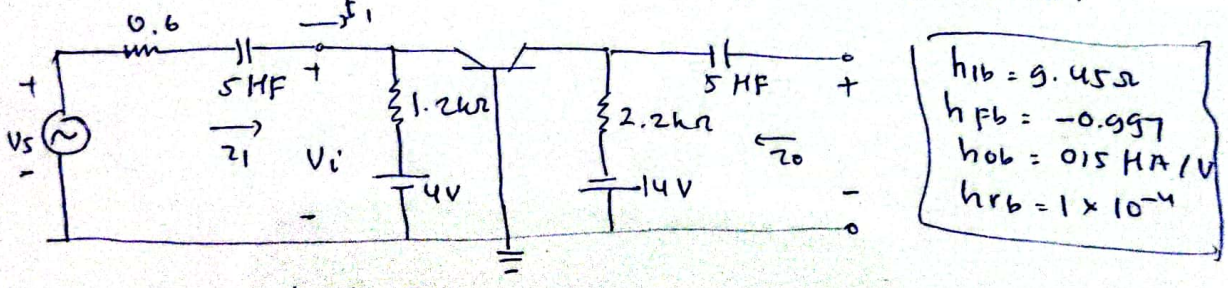
$= (132.70)(0.998)$ $\frac{I_i}{I_i'} = 0.998$

$= 132.43$

d). $Z_o = \frac{1}{h_{oe} - (h_{fe}h_{re} / (h_{ie} + R_s} = \frac{1}{(25\mu S)((140)(1.5\times10^{-4})) (0.86k\Omega + 1k\Omega}$

$= \frac{1}{13.71\mu S} = 72.9k\Omega$

26 for the common-base amplifier of fig. 8.84, determine:



a). $Z_i' = h_{ib} - \frac{h_{fb} h_{rb} R_L}{1 + h_{ob} R_L} = 9.45k\Omega - \frac{(-0.997)(1\times10^{-4})(2.2k\Omega)}{1 + (0.5\,H\frac{A}{V})(2.2k\Omega)} = 9.67\Omega$

$Z_i = 1.2k\Omega \,//\, Z_i' = 1.2k\Omega \,//\, 9.67\Omega = 9.59\Omega$

b). $A'i = \frac{h_{fb}}{1 + h_{ob} R_L} = \frac{0,997}{1 + (0.5\,H\frac{A}{V})(2.2k\Omega)} = -0,996$

$A_i = \frac{I_o}{I_i'} \cdot \frac{I_i'}{I_i} = (-0.996)\frac{1.2k\Omega}{1.2k\Omega + 9.67k\Omega} = -0.998$

c). $A_V = \frac{-h_{fb} R_L}{h_{ib} + (h_{ib}h_{ob} - h_{fb}h_{fb}) R_L} = \frac{-(-0.997)(2.2k\Omega)}{9.45\Omega + ((945\Omega))(0.5\,HA/V) - (0.997(1\times10^{-4}))\,2.2k\Omega}$

$= 226.21$

d). $Z'o = \frac{1}{h_{ob} - (h_{fb}h_{rb} / h_{ib})} = \frac{1}{(0.5\,HA/V) - ((-0.997)(1\times10^{-4}))/9.45\Omega} = 905k\Omega$

$Z_o = 2.2k \,//\, Z_o' = 2.15k\Omega$

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